

Protocol Status Memo

Date: 07-MAR-2023

Principal Investigator: Kirby Nielsen

Title of Project: Expected Utility Violations and Decision Error

Caltech IRB Number: IR23-1289

Type of Application: Initial Query

Funding Agency: CIT ENDOWMENT

Review Type: Chair Review

IRB Decision: Exempt

Exemption Type: Educational tests, survey, interview, observation of public behavior (45 C.F.R. § 46.104(d)(2)(i),(ii))

Approval Date: 17-MAR-2023

Risk Type: Minimal Risk

Maximum Number of Participants Approved:

Summary of Project:

The expected utility model is the workhorse model of choice under uncertainty. Many experiments have documented violations of expected utility. We hypothesize that decisions are prone to error, and many violations of expected utility can be attributed to decision error rather than non-standard preferences. We introduce a new methodology that is robust to decision error and hypothesize that this new methodology will result in fewer violations of expected utility. Using this new methodology, we will run a lottery choice experiment in which individuals make binary lottery choices that allow us to test expected utility.

Dear Kirby Nielsen,

This project/research was received, reviewed and determined to be **Exempt** by the Caltech Committee for the Protection of Human Subjects (IRB) on 07-MAR-2023 and ratified by the Committee at the 17-MAR-2023 meeting.

This project meets the criteria for exemption from further IRB review, pursuant to Educational tests, survey, interview, observation of public behavior (45 C.F.R. § 46.104(d)(2)(i),(ii)), which states:

(45 C.F.R. § 46.104(d)(2)(i),(ii)): **Research that only includes interactions involving educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures or observation of public behavior (not including visual or auditory recording), if at least one of the following criteria are met:** (i) the information obtained is recorded in such a manner that the identity of the subjects cannot be readily ascertained, directly or through identifiers linked to the subjects; (ii) any disclosure of the subjects' responses outside the research would not reasonably place the subjects at risk of criminal or civil liability or be damaging to

the subjects' financial standing, employability, educational advancement, or reputation.

If you wish to change any aspect of this exempt study including, but not limited to, study procedures, consent forms, investigators, funding, or advertisements, you are required to submit a query amendment and obtain a new IRB exemption and/or approval prior to implementation of these changes. Note that some changes may change the nature of your protocol from exempt to non-exempt.

Adverse or unanticipated events must be reported immediately to the IRB Office at irb@caltech.edu, or 626-395-8448.

You can refer to the Caltech Institutional Review Board Policy, found online at https://researchcompliance.caltech.edu/documents/22048/Caltech_IRB_Policy.pdf. Should you have any questions regarding your protocol or your IRB approval, please contact the IRB Administrator at irb@caltech.edu, 626-395-8448, or the Office of Research Compliance at orc@caltech.edu.

This review was carried out in accordance with the requirements of Part 46, "Protection of Human Subjects" of Title 45 of the Code of *Federal Regulations* and the "U.S. Department of Health and Human Services (DHHS) Federal-Wide Assurance (FWA) for the Protection of Human Subjects for Domestic Institutions," California Institute of Technology Assurance # FWA00003897.

Signature: 
Ralph Adolphs
Chairman, Committee for the Protection of Human Subjects
Date: 07-MAR-2023